

Encoding Schemes and Number System

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'A' = 65 = 0100 0001
↑ ↑ ↑
letter code Binary Number
(Decimal value)

Encoding of data entered using keyboard.

'ज' on hindi keyboard hex value 0905
whose binary equivalent is 0000 1001 0000 0101

What is encoding?

The mechanism of converting data into an equivalent cipher using specific code is called encoding.

It is important to understand why code value 65 is used for key 'A' and not any other value? It is same for all the keyboards, irrespective of their make?

Yes, it is same for all the keyboards. This has been possible because of standard encoding scheme, where each letter, numerals and symbols is encoded or assigned a unique code. Some of the well-known encoding schemes are:

1. ASCII (American Standard Code for Information Interchange).

7-bit ASCII code.

$2^7 = 128$

@ → 64

A → 65

⋮

Z → 90

a → 97

z → 122

spacebar → 32 ['a' - 'A']

0 → 48, 9 → 57.

! → 33 ' → 34 # → 35 \$ → 36
 % → 37 & → 38 (→ 40) → 41
 ' → 39

UNICODE

Encoding schemes for character sets of different languages. But they were not able to communicate with each other as each of them represents character in their own ways. Hence, text created using one encoding scheme was not recognized by another machine using the different encoding scheme.

So therefore a standard. Called Unicode, has been developed to incorporate all the characters of every written language in of the world. Unicode provides a unique number for every character, irrespective of device (server, desktop, mobile) Operating system (Linux, Windows, IOS) or software applications (different browsers, text editors, etc). Commonly used Unicode encodings are UTF-8, UTF-16 and UTF-32. It is a super set of ASCII and the values zero - 128 have the same character as in ASCII. Unicode characters for Devnagari script is shown in table below.

Devanagari

Showing 120 of 128

ि U+0900	ी U+0901	ं U+0902	ः U+0903	ऐ U+0904	अ U+0905	आ U+0906	इ U+0907
ई U+0908	उ U+0909	ऊ U+090A	ऋ U+090B	लृ U+090C	एँ U+090D	ऐँ U+090E	ए U+090F
ऐ U+0910	ऑ U+0911	ओ U+0912	ओ U+0913	औ U+0914	क U+0915	ख U+0916	ग U+0917
घ U+0918	ङ U+0919	च U+091A	छ U+091B	ज U+091C	झ U+091D	ञ U+091E	ट U+091F
ठ U+0920	ड U+0921	ढ U+0922	ण U+0923	त U+0924	थ U+0925	द U+0926	ध U+0927

न U+0928	न U+0929	प U+092A	फ U+092B	ब U+092C	भ U+092D	म U+092E	य U+092F
र U+0930	र U+0931	ल U+0932	ळ U+0933	ळ U+0934	व U+0935	श U+0936	ष U+0937
स U+0938	ह U+0939	ं U+093A	ी U+093B	ृ U+093C	ऽ U+093D	ा U+093E	ि U+093F
ी U+0940	ु U+0941	ू U+0942	ृ U+0943	ृ U+0944	ँ U+0945	ै U+0946	े U+0947
ै U+0948	ॉ U+0949	ो U+094A	ो U+094B	ौ U+094C	र् U+094D	ि U+094E	ौ U+094F
ॐ U+0950	ं U+0951	ं U+0952	ं U+0953	ं U+0954	ँ U+0955	ँ U+0956	ँ U+0957
क U+0958	ख U+0959	ग U+095A	ज U+095B	ड़ U+095C	ढ़ U+095D	फ़ U+095E	य़ U+095F
ऋ U+0960	ॠ U+0961	ॡ U+0962	ॢ U+0963	। U+0964	॥ U+0965	० U+0966	१ U+0967
२ U+0968	३ U+0969	४ U+096A	५ U+096B	६ U+096C	७ U+096D	८ U+096E	९ U+096F
० U+0970	१ U+0971	अ U+0972	अ U+0973	आ U+0974	औ U+0975	अ U+0976	अ U+0977
रु U+0978	ज़ U+0979	ष U+097A	ग़ U+097B	ज़ U+097C	१ U+097D	ड U+097E	ब U+097F

Number System

Every number system has a set of unique characters or literals. The count of these literals is called the radix or base of the number system. The four different number system used in this context of computer.

Binary Base: 2 Comprises: 0, 1
Decimal Base: 10 Comprises: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Hexadecimal Base: 16 Comprises: 0-9 and A-F
Octal Base: 8 Comprises: 0, 1, 2, 3, 4, 5, 6, 7

Number system
for
Computers

Four different number systems.

Integer part fractional part ← real numbers

Digit	1	2	3	.	4	5
Positional number	2	1	0		-1	-2
Position value	10^2	10^1	10^0		10^{-1}	10^{-2}

$$1 \times 10^2 + 2 \times 10^1 + 3 \times 10^0 + 4 \times 1/10 + 5 \times 1/100$$
$$100 + 20 + 3 + .25 + 0.2 = (123.45)_{10}$$

Computation of decimal number using its positional value.

Binary Number System

Digit	1	0	1	1	.	1	0	1
Positional number	3	2	1	0		-1	-2	-3
Positional value	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}

$$1 * 8 + 1 * 2 + 1 * 1 + 1/2 + 1/8 = 11.625$$

Octal number system: Octal number system is called base - 8 system. As its

total digits (0 - 7) And positional value is expressed in powers of 8.

Digit	2	4	7	.	2	4
Positional number	2	1	0		-1	-2
Positional value	8^2	8^1	8^0		8^{-1}	8^{-2}

$2 * 8^2 + 4 * 8 + 7 * 1 + 2 * 1/8 + 4/64 = 167.3125$ (This is decimal number)

Hexadecimal number system:

Hexadecimal numbers are also used for compact representation of binary numbers. It consists of 16 unique symbols (0 - 9, A - F) and is called the base - 16 system. In hexadecimal system, each alphanumeric digit is represented as a group of four binary digits because four bits ($2^4 = 16$) Are sufficient to represent 16 alphanumeric symbols.

Digit	2	3	A	.	0	5
Positional number	2	1	0		-1	-2
Positional value	256	16	1		1/16	1/256

$256*2 + 16 * 3 + 10 * 1 + 5/256 = 570.0195$ (Decimal Value)

Applications of hexadecimal number system.

- Main memory is made up of memory locations where each location has a unique address. Usually size of a memory address is 16 bit or 32 bit. To access 16 bit memory address, a programmer has to use 16 binary bits, which is difficult to deal with. To simplify the address representation, hexadecimal and octal numbers are used. Let us consider for a 16 bit memory address 1100000011110001. Using the hexadecimal notation, this address is mapped to C0F1. Which is more easy to remember. The equivalent octal representation for 16 - bit value is **140361**.

Python code:

```
print(oct(0b1100000011110001))  
0o140361
```

- Hexadecimal numbers are also used for describing the colors on the web page. Each color is made up of three primary colors, red, green and blue, popularly called RGB, in short. In most color maps, each color is usually chosen from the palette of 16 million colors. Therefore, 24 bits are required for representing each color having three components. (8 bits were read, 8 bits for green, eight bits for blue component) It is difficult to remember 24 - bit binary color code. Therefore, color codes are written in hexadecimal form for compact representation. For example

24 - bit code for red color is. 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0. The equivalent hexadecimal notation is (FF,00,00), which can be easily remembered.